

Statistics for Scientific Claims

Unit 2

Descriptive Statistics and Visualisation

Lecture notes for IB DP Biology and Chemistry

Course purpose	To help students move from raw experimental data to cautious, justified scientific claims.
Unit focus	Summaries, spread, error bars, raw-data plots, histograms, box plots, scatter plots, and honest visual evidence.
Central habit	Before running an inferential test, describe the data clearly and show enough of the raw pattern for the reader to judge it.

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1 Purpose of Unit 2

Unit 1 argued that statistics begins with measurement quality and data type. Unit 2 moves to the next step: describing and visualising data before formal hypothesis testing.

Students often rush toward a test because software can return a p-value quickly. That is poor practice. Before testing, a scientist should know the basic shape of the data: centre, spread, outliers, grouping, possible skew, and whether the pattern is visible in the raw observations.

Core thesis

Descriptive statistics and graphs do not merely decorate a report. They are the first test of whether a scientific claim is even plausible.

By the end of this unit, students should be able to:

1. calculate and interpret mean, median, range, interquartile range, variance, standard deviation, standard error, and simple confidence intervals;
2. choose appropriate graphs for categorical, count, and continuous data;
3. distinguish SD, SEM, and confidence intervals as different kinds of error bars;
4. explain why bar charts of means often hide important information;
5. use dot plots, box plots, histograms, and scatter plots responsibly;
6. write descriptive conclusions before moving to formal inference.

2 Description before inference

A good statistical workflow starts with description:



The graph and summary should come before the test. If a graph shows a strong outlier, a curved relationship, unequal spread, or a ceiling effect, the planned test may no longer be appropriate.

Common failure

A student calculates a p-value without ever plotting the data. This is dangerous because the test may be answering a question that does not match the actual pattern.

3 A running dataset: plant height

We will use a small plant-growth dataset throughout the unit. Five plants were grown in a control condition and five with fertilizer.

Group	Plant 1	Plant 2	Plant 3	Plant 4	Plant 5
Control height / cm	11.8	12.6	12.1	13.0	12.5
Fertilizer height / cm	14.2	15.8	13.9	16.4	15.2

A weak statement would be:

The fertilizer worked because the fertilizer plants were taller.

A better descriptive beginning is:

The fertilizer group had a higher mean height, but it also showed greater spread. Because the sample size is small, the difference should be interpreted cautiously and shown with the raw data.

4 Measures of centre

A measure of centre describes a typical value.

4.1 Mean

The arithmetic mean is:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum x_i}{n}.$$

For the control group:

$$\bar{x} = \frac{11.8 + 12.6 + 12.1 + 13.0 + 12.5}{5} = 12.4 \text{ cm.}$$

For the fertilizer group:

$$\bar{x} = \frac{14.2 + 15.8 + 13.9 + 16.4 + 15.2}{5} = 15.1 \text{ cm.}$$

The mean uses every value, so it responds strongly to extreme values.

4.2 Median

The median is the middle value after sorting the data.

Control sorted:

11.8, 12.1, 12.5, 12.6, 13.0.

The median is:

12.5 cm.

Fertilizer sorted:

13.9, 14.2, 15.2, 15.8, 16.4.

The median is:

$$15.2 \text{ cm.}$$

The median is less sensitive to extreme values than the mean. It is often better for strongly skewed data.

4.3 Mode

The mode is the most frequent value. It is useful for categorical data and some discrete count data, but it is often not very useful for continuous measurements because exact repeated values may be rare.

Choosing a centre

Use the mean when the distribution is roughly symmetric and extreme values are not dominant.
Use the median when the data are skewed or when outliers strongly affect the mean.

5 Measures of spread

A measure of spread describes how variable the data are. Two groups can have the same mean but very different spread.

5.1 Range

The range is:

$$\text{range} = \text{maximum} - \text{minimum.}$$

For the control group:

$$13.0 - 11.8 = 1.2 \text{ cm.}$$

For the fertilizer group:

$$16.4 - 13.9 = 2.5 \text{ cm.}$$

The range is easy to understand, but it depends only on the two most extreme values.

5.2 Interquartile range

The interquartile range, or IQR, is:

$$\text{IQR} = Q_3 - Q_1.$$

It describes the spread of the middle 50% of the data. It is less sensitive to outliers than the range.

For the control group, using the quartiles from the sorted sample:

$$Q_1 = 12.1, \quad Q_3 = 12.6, \quad \text{IQR} = 0.5 \text{ cm.}$$

For the fertilizer group:

$$Q_1 = 14.2, \quad Q_3 = 15.8, \quad \text{IQR} = 1.6 \text{ cm.}$$

Different software may use slightly different quartile conventions. For school reports, the convention is less important than being consistent and transparent.

5.3 Sample variance and sample standard deviation

The sample variance is:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}.$$

The sample standard deviation is:

$$s = \sqrt{s^2}.$$

The standard deviation describes the typical spread of individual observations around the mean.

For the running dataset:

Group	Mean / cm	Median / cm	Range / cm	IQR / cm	Sample SD / cm
Control	12.4	12.5	1.2	0.5	0.46
Fertilizer	15.1	15.2	2.5	1.6	1.05

Interpretation

The fertilizer group had a higher centre and a larger spread. A graph should therefore show both facts: not only that the group average is higher, but also that the fertilizer group values are more variable.

6 Standard error and confidence interval

Standard deviation and standard error are often confused.

6.1 Standard deviation

SD describes the spread of individual observations:

$$s = \text{spread of individual data points around the sample mean.}$$

If plant heights have a large SD, individual plants are very different from each other.

6.2 Standard error of the mean

The standard error of the mean, or SEM, is:

$$\text{SEM} = \frac{s}{\sqrt{n}}.$$

SEM describes the uncertainty in the estimated mean, not the spread of individual observations.

For the control group:

$$\text{SEM} = \frac{0.46}{\sqrt{5}} \approx 0.21 \text{ cm.}$$

For the fertilizer group:

$$\text{SEM} = \frac{1.05}{\sqrt{5}} \approx 0.47 \text{ cm.}$$

6.3 Simple confidence interval for a mean

For small samples, a simple 95% confidence interval for the mean is:

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}.$$

For the plant dataset, with $n = 5$, the degrees of freedom are 4, and $t^* \approx 2.776$ for a 95% interval.

Control:

$$12.4 \pm 2.776(0.21) \approx 12.4 \pm 0.58.$$

So the interval is approximately:

$$11.82 \text{ to } 12.98 \text{ cm.}$$

Fertilizer:

$$15.1 \pm 2.776(0.47) \approx 15.1 \pm 1.31.$$

So the interval is approximately:

$$13.79 \text{ to } 16.41 \text{ cm.}$$

Do not confuse SD, SEM, and CI

SD shows spread among individual observations. SEM shows uncertainty in the estimated mean. A confidence interval shows a plausible range for the population mean under the model assumptions.

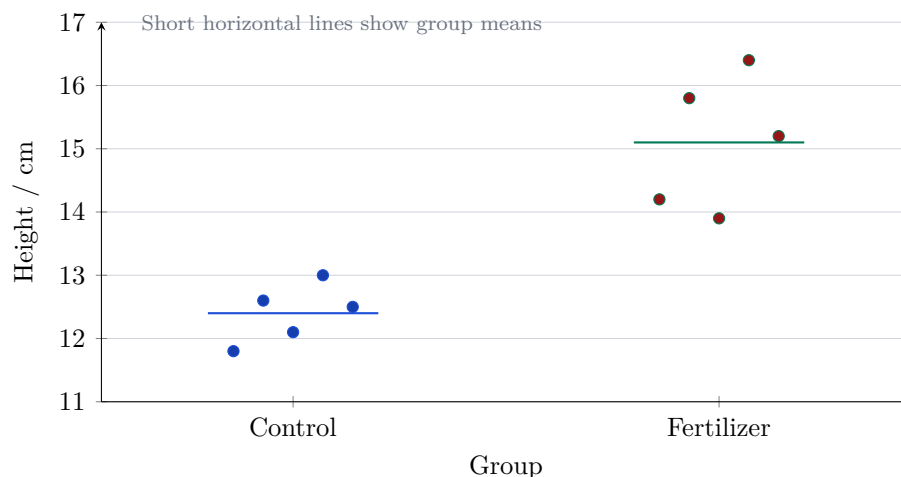
Quantity	What it describes	Good use
Range	Distance from minimum to maximum	Quick rough spread, especially for small classroom datasets
IQR	Spread of middle 50%	Skewed data or box plots
SD	Spread of individual values	Variation within a group
SEM	Uncertainty in the sample mean	Precision of the estimated mean
95% CI	Plausible range for population mean	Scientific reporting and comparison of estimates

7 Raw-data plots should usually come first

With small or moderate IB datasets, raw data should usually be visible. A plot of raw points lets the reader see sample size, spread, outliers, and overlap between groups.

7.1 Dot plot

A dot plot is often the best first graph for small datasets.



This plot shows that fertilizer values are generally higher, but it also shows the small sample size and the larger spread in the fertilizer group.

Bar charts hide too much

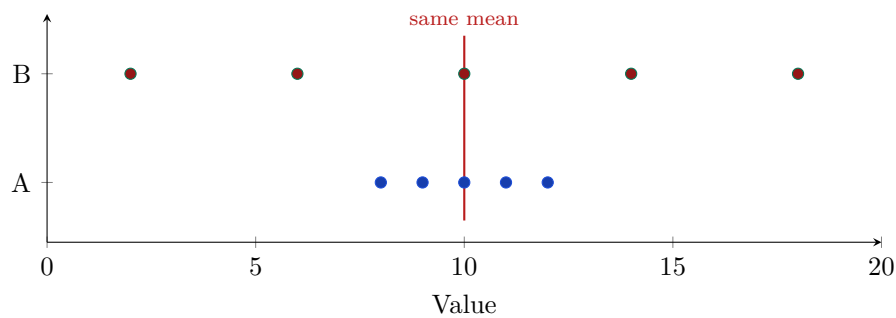
A bar chart of two means would hide the fact that each group has only five plants. It would also hide the greater spread in the fertilizer group.

7.2 Same mean, different spread

Two datasets can have the same mean but very different variation:

Dataset	Value 1	Value 2	Value 3	Value 4	Value 5
A	8	9	10	11	12
B	2	6	10	14	18

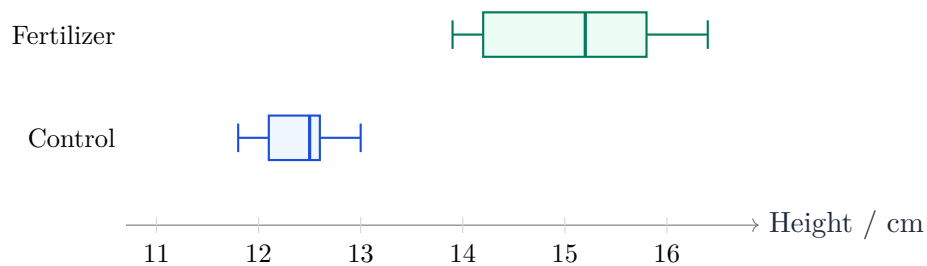
Both means are 10, but dataset B is much more variable.



A mean alone is therefore incomplete. It must be paired with a measure of spread and, where possible, a raw-data plot.

8 Box plots

A box plot summarises median, quartiles, range, and possible outliers. It is useful when there are enough observations to make quartiles meaningful. For very small samples, dot plots are usually more informative.



Box plot interpretation

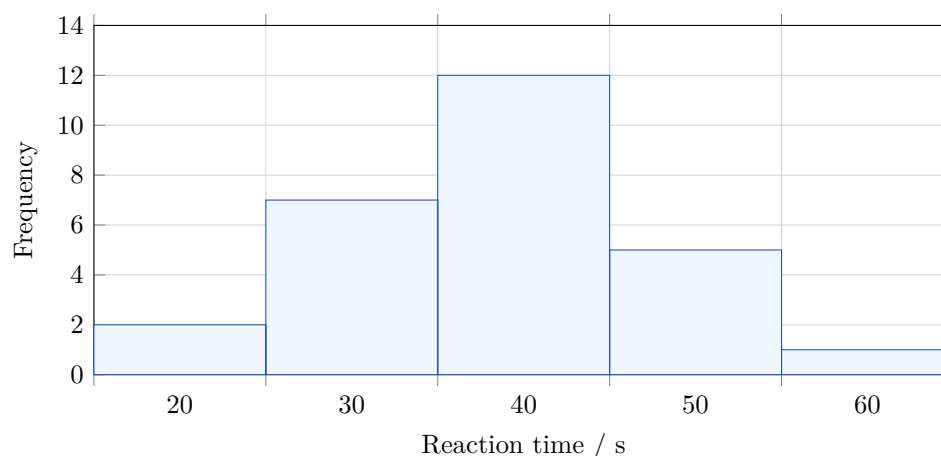
The line inside the box is the median. The box shows the middle 50% of the data. The whiskers show the lower and upper extremes, unless a special outlier rule is used.

9 Histograms

A histogram shows the shape of a distribution for continuous or near-continuous measurements. It groups values into intervals called bins.

Suppose a class records reaction times for an enzyme colour-change endpoint:

Time interval / s	20–29	30–39	40–49	50–59	60–69
Frequency	2	7	12	5	1



The histogram suggests that values cluster around 40–49 seconds, with fewer very short or very long times.

Bin choice matters

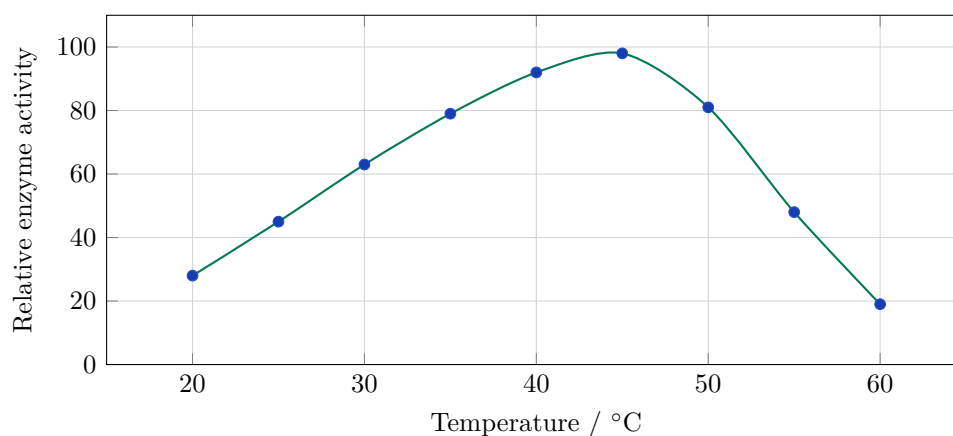
Changing bin width can change the visual impression. Do not choose bins to exaggerate a desired pattern. For small datasets, a dot plot may be more honest than a histogram.

10 Scatter plots

A scatter plot is used when both variables are numerical and the question concerns a relationship.

Example research question:

How does temperature affect enzyme activity between 20°C and 60°C?



This pattern is not linear across the whole range. A straight-line regression would be inappropriate for the full dataset. A descriptive graph prevents a bad statistical model.

Interpretation

The scatter plot suggests enzyme activity increases from 20°C to about 45°C, then decreases sharply. A single linear correlation would hide the biological interpretation because enzyme activity is expected to have an optimum temperature.

11 Categorical data: counts before percentages

For categorical outcomes, show counts first and percentages second.

Example: seed germination under control and salt conditions.

Condition	Germinated	Not germinated	Germination percentage
Control	18	2	90%
Salt solution	12	8	60%

A clear descriptive statement is:

The salt group had lower germination in this sample: 12 of 20 seeds germinated compared with 18 of 20 in the control group. This corresponds to a difference of 30 percentage points.

Percentages without counts are weak evidence

A difference between 90% and 60% is more informative if the reader knows whether it came from 10 seeds, 20 seeds, or 1000 seeds. Always show counts.

12 Error bars

Error bars are not all the same. A graph must state what the error bars represent.

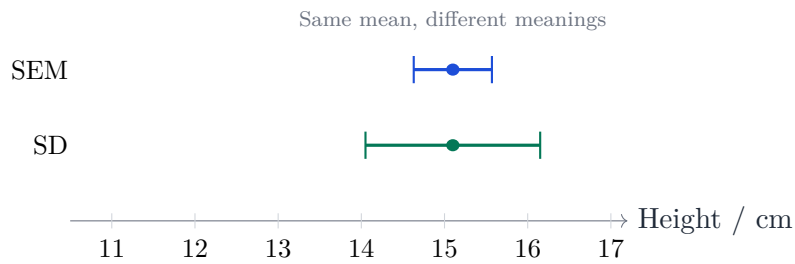
Error bar	Meaning	Common misuse
SD	Variation among individual observations	Used as if it were uncertainty in the mean
SEM	Precision of the sample mean estimate	Used to make variation look artificially small
95% CI	Plausible range for the population mean under assumptions	Treated as proof that the true mean is inside the interval
Instrument uncertainty	Measurement resolution/tolerance	Used while ignoring biological variation

12.1 Visual comparison of SD and SEM

For the plant dataset, the fertilizer group has:

$$\bar{x} = 15.1, \quad s = 1.05, \quad \text{SEM} = 0.47.$$

SEM is smaller than SD because it describes uncertainty in the mean, not variation among plants.



Reporting rule

Every graph with error bars must label the error bar type: SD, SEM, CI, instrument uncertainty, or another defined quantity.

13 Choosing the right graph

Data or question	Better graph	Avoid or use cautiously
Small continuous group data	Dot plot with mean/median line	Bar chart of means only
Larger continuous distribution	Histogram, density plot, box plot	Too many bins or too few bins
Two numerical variables	Scatter plot	Connecting points if there is no ordered sequence
Categorical counts	Bar chart of counts, contingency table	Percentages without sample size
Time sequence	Line graph, with raw points if possible	Smoothing without explanation
Before/after same samples	Paired dot plot or line plot	Independent-group bar chart

14 Worked example 1: descriptive report for plant height

14.1 Raw data

Group	1	2	3	4	5
Control / cm	11.8	12.6	12.1	13.0	12.5
Fertilizer / cm	14.2	15.8	13.9	16.4	15.2

14.2 Summary table

Group	n	Mean	Median	SD	SEM	Range
Control	5	12.4	12.5	0.46	0.21	1.2
Fertilizer	5	15.1	15.2	1.05	0.47	2.5

14.3 Descriptive conclusion

The fertilizer group had a higher mean plant height than the control group by 2.7 cm. The fertilizer group also had greater spread, with an SD of 1.05 cm compared with 0.46 cm in the control group. Because both groups contain only five plants, the pattern should be shown with raw data and interpreted cautiously before formal testing.

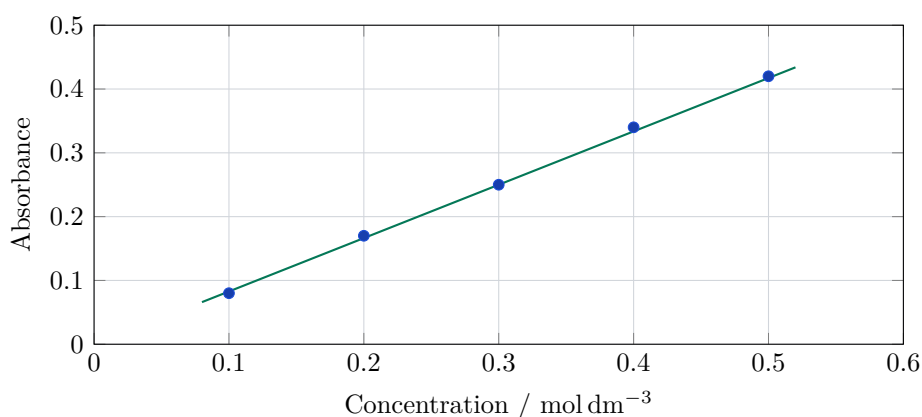
This is a descriptive conclusion. It does not yet claim that the effect is statistically significant or caused by fertilizer. That requires later units on experimental design and inference.

15 Worked example 2: choosing a graph

A student investigates copper sulfate concentration using a colorimeter. The data are:

Concentration / mol dm^{-3}	Absorbance
0.10	0.08
0.20	0.17
0.30	0.25
0.40	0.34
0.50	0.42

Both variables are numerical. The correct first graph is a scatter plot.



A responsible descriptive statement:

Absorbance increased approximately linearly over the tested concentration range. The scatter plot suggests that a linear calibration model may be reasonable within this range, but this does not justify extrapolation beyond the measured standards.

16 Worked example 3: categorical counts

A student reports:

90% of control seeds germinated and 60% of salt-treated seeds germinated.

This is incomplete because sample size is hidden. The stronger version is:

18 out of 20 control seeds germinated, compared with 12 out of 20 salt-treated seeds. The germination percentage was therefore 90% in the control condition and 60% in the salt condition, a difference of 30 percentage points.

For categorical count data, a table of counts is often more important than a graph.

17 Language for descriptive statistics

Weak phrase	Better phrase
The average proves the treatment worked.	The treatment group had a higher mean in this sample.
The SD proves the results are accurate.	The SD describes spread, not accuracy.
The SEM shows the variation in the data.	The SEM shows uncertainty in the estimated mean; SD shows variation among observations.
The graph shows a perfect relationship.	The scatter plot suggests an approximately linear relationship within the measured range.
The bar chart shows the result clearly.	The bar chart hides sample size and spread; raw points should be shown when possible.
The data are reliable because the mean is high.	Reliability concerns consistency, measurement quality, and replication, not whether the mean is high.

Model sentence

The data had a mean of ... and a standard deviation of ... The graph suggests ..., but the small sample size, spread, and possible outliers mean that the conclusion should be cautious.

18 Mini-practice tasks

Task 1: Centre and spread

The reaction times for five trials are:

32, 35, 36, 39, 48 s.

Calculate the mean, median, range, and comment on whether the largest value may affect the mean.

Task 2: SD or SEM?

A graph compares the mean enzyme activity of two groups. The caption says only “error bars show error.” Explain why this is unacceptable. State what should be written instead.

Task 3: Choose the graph

For each investigation, choose a suitable graph.

- Plant height in control and fertilizer groups, with $n = 6$ per group.
- Absorbance measured at five known concentrations.
- Number of bacterial colonies per plate across 40 plates.
- Survival counts under control and treatment conditions.

Task 4: Counts and percentages

A student writes: “The survival rate was 75% in group A and 50% in group B.” What information is missing? Rewrite the sentence assuming group A had 15 survivors out of 20 and group B had 10 survivors out of 20.

Task 5: Better descriptive conclusion

Rewrite this claim more cautiously:

The fertilizer increased plant growth because the mean was higher.

Use the following statistics: control mean 12.4 cm, fertilizer mean 15.1 cm, control SD 0.46 cm, fertilizer SD 1.05 cm, $n = 5$ per group.

19 Answer key

Task 1

Mean:

$$\bar{x} = \frac{32 + 35 + 36 + 39 + 48}{5} = 38 \text{ s.}$$

Median: sorted values are already 32, 35, 36, 39, 48, so the median is 36 s.

Range:

$$48 - 32 = 16 \text{ s.}$$

The value 48 s is much larger than the others and pulls the mean upward. It should not be deleted automatically, but it should be shown in the raw data and discussed.

Task 2

The phrase “error bars show error” is unacceptable because different error bars mean different things. The caption should specify SD, SEM, 95% confidence interval, instrument uncertainty, or another defined quantity. For example: “Points show group means; error bars show ± 1 SD.”

Task 3

- Dot plot with group means or medians. A box plot may be added, but raw points are important because $n = 6$ is small.
- Scatter plot, possibly with a fitted calibration line if the relationship is approximately linear.
- Histogram or dot plot of colony counts; if comparing conditions, separate panels or grouped dot plots.

d) Count table and bar chart of counts or proportions, with sample sizes shown.

Task 4

The sample sizes and counts are missing. A better sentence is:

In group A, 15 out of 20 organisms survived, giving a survival rate of 75%. In group B, 10 out of 20 survived, giving a survival rate of 50%. The observed difference was 25 percentage points.

Task 5

Possible rewrite:

The fertilizer group had a higher mean plant height than the control group in this sample: 15.1 cm compared with 12.4 cm, a difference of 2.7 cm. The fertilizer group also had greater spread, with SD 1.05 cm compared with 0.46 cm for the control group. Because the sample size was small, the result should be shown with raw data and interpreted cautiously before formal inference.

20 Unit 2 summary

What students should carry forward

Before running a statistical test, ask:

1. What is the data type?
2. What is the centre of the data?
3. How spread out are the observations?
4. Are there outliers or skew?
5. Does the graph show the raw pattern clearly?
6. Are error bars labelled correctly?
7. What descriptive claim is justified before formal inference?

The next unit should build from this foundation into distributions and data-generating processes: normal, binomial, Poisson, log-normal, t, chi-square, F, and sampling distributions.